Innate Cognitive Abilities in Newborns

In cognitive psychology and neuroscience, the innate abilities that a child possesses at birth are primarily defined as **Core Knowledge systems** (or core knowledge), or more generally, **innate predispositions** and **neonatal competencies**.

These concepts move beyond the traditional view of the newborn as a *tabula rasa* (blank slate), demonstrating that the human brain arrives already equipped with specific information packages and learning mechanisms.

Main Terms and Concepts

• **Core Knowledge (Core Knowledge Systems):** Theorized by cognitive scientist Elizabeth Spelke, this term refers to four (or five) innate cognitive systems that allow newborns to understand the physical and social world. These include an innate understanding of **objects** (intuitive physics), **numbers** (number sense), **space** (geocentric orientation), and **social agents** (understanding others' intentions).

• **Biological or Innate Predispositions:** These refer to the structural constraints of the brain that direct the newborn's attention toward biologically relevant stimuli, such as the human face (a phenomenon known as *conspecific* orientation).

• **Neonatal (or Archaic) Reflexes:** At a more strictly neurological level, these are automatic and involuntary motor responses (such as the sucking or grasping reflexes) controlled by subcortical brain structures.

• **Modular Innatism:** A concept introduced by Jerry Fodor and partially revised by developmental cognitive neuroscience, suggesting that the mind is born already divided into specialized 'modules' to process specific types of information (e.g., language).

Academic Bibliography

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